PHYSICAL CUMULATIVE EXAM

October 22, 1994

A number of different techniques have been developed for studying the kinetics of fast reactions. Among these techniques are the stopped-flow method and relaxation method. Describe, in general terms, the principle of the stopped-flow method.

If the half-life of a reaction is less than 10^{-3} sec then the stopped-flow method cannot be used. To extend kinetic methods to the study of reactions whose half-lives are of the order of 10^{-6} sec or less, Manfred Eigen developed the relaxation method. This method involves a system initially at equilibrium under a given set of experimental conditions. These conditions are then suddenly changed, the system is no longer at equilibrium, and the system relaxes back to a state of equilibrium. The rate at which it relaxes can be measured and the rate constants can be determined. For the hypothetical reaction:

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

derive the equation that relates the relaxation time, τ, to the rate constant for the forward reaction, k_1, and the rate constant for the reverse reaction, k_{-1}. Discuss how you would determine these rate constants experimentally using the relaxation method.